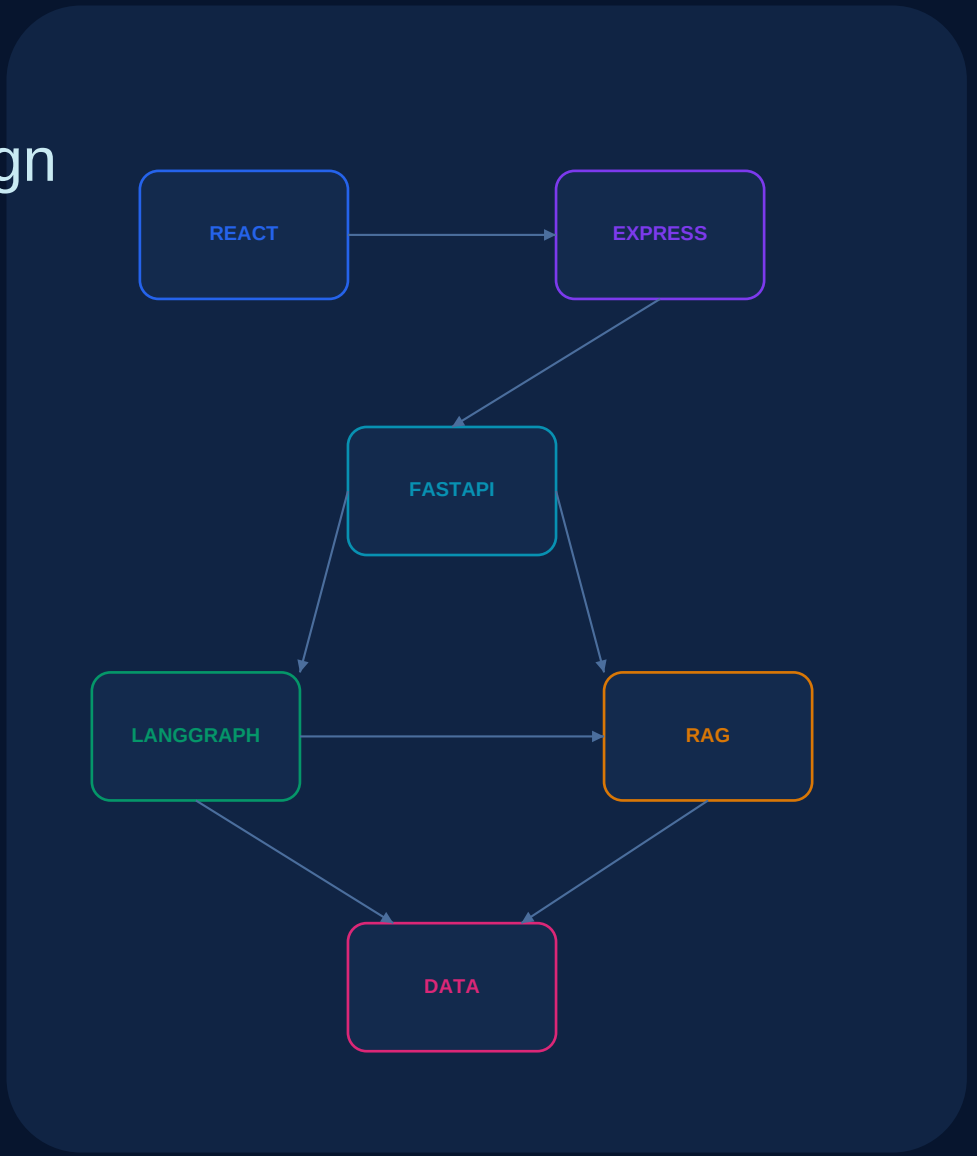
SYSTEM DESIGN DOCUMENT

LawShield

High-Level Design + Low-Level Design

A code-derived architecture reference for the Indian legal AI platform: service boundaries, data ownership, multi-agent execution, document RAG, privacy, APIs, persistence and operating constraints.



Document scope

- HLD: context, containers, data stores, security, deployment and reliability.
- LLD: modules, state machines, APIs, schemas, algorithms and runtime limits.
- Implementation status: current behavior plus clearly identified design gaps.

Repository snapshot: 28 September 2026

Purpose, scope and architecture context

CONTEXT

Primary actors

- Authenticated citizen or legal professional using the browser workspace.
- Administrator/operator running application and data infrastructure.
- External LLM providers used behind a pluggable provider abstraction.

VALUE

System outcomes

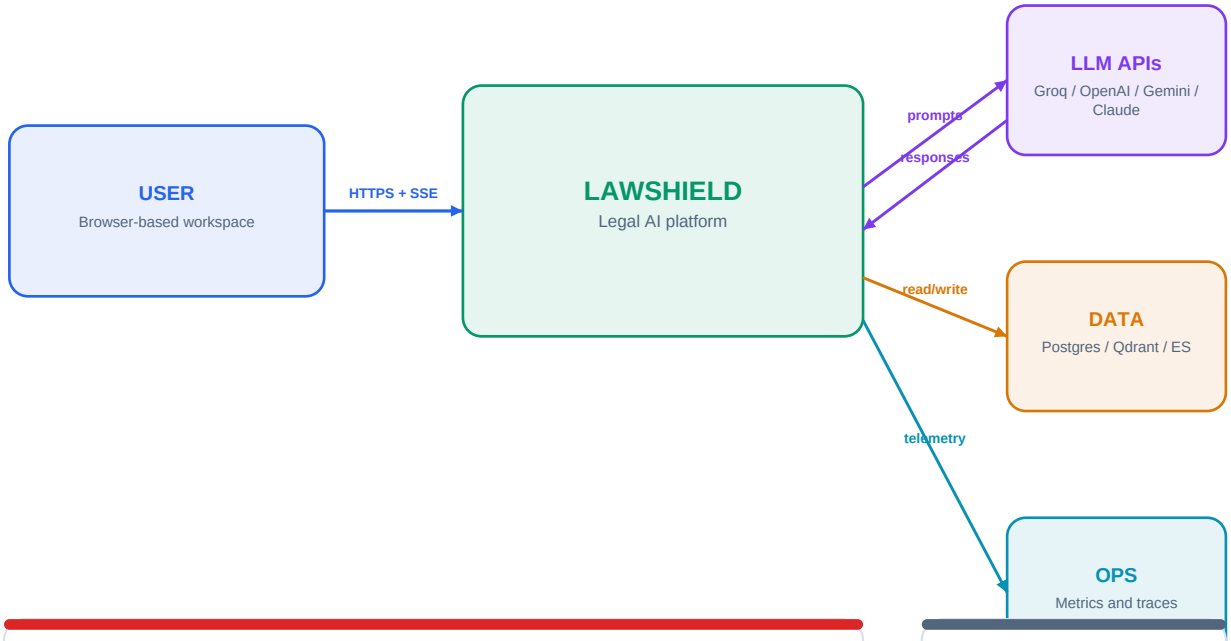
- Conversational Indian-law assistance with specialist routing.
- Document-grounded answers with hybrid retrieval and citations.
- PII masking before model invocation and restoration after validation.

DRIVERS

Quality attributes

- Grounding and bounded retries over unbounded autonomous loops.
- Durable ownership in PostgreSQL; fast semantic and lexical evidence stores.
- Progressive SSE feedback and graceful dependency fallbacks.

C4-style system context



IMPORTANT

Trust boundary

- Browser never calls an LLM provider directly.
- Express validates identity and resource ownership before AI requests.
- FastAPI masks PII before LangGraph or model execution.

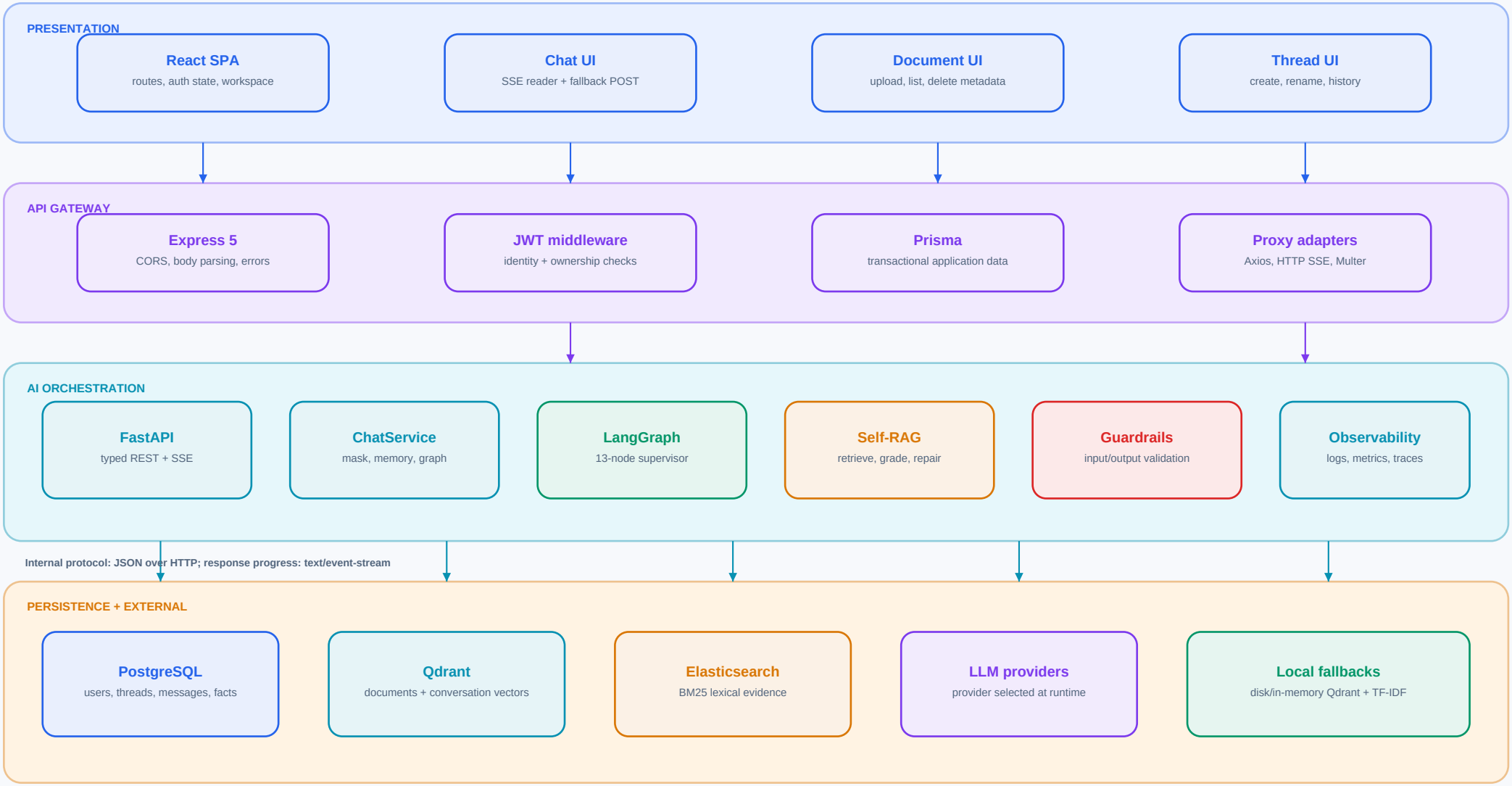
CURRENT

Out of scope

Multi-tenant namespace, Kubernetes deployment, billing or human case-management workflow is implemented.

Logical architecture and service boundaries

Three application services with purpose-specific persistence



Data architecture, ownership and consistency

One durable source of truth plus two retrieval-optimized indexes

SYSTEM OF RECORD

PostgreSQL

- Durable entities: User, Thread, Message, LongTermFact and Document.
- User owns threads and document metadata; thread owns messages and facts.
- Cascade delete is configured for thread messages/facts and user relations.
- Prisma serves Node routes; psycopg2 serves AI long-term fact memory.

SEMANTIC STORE

Qdrant

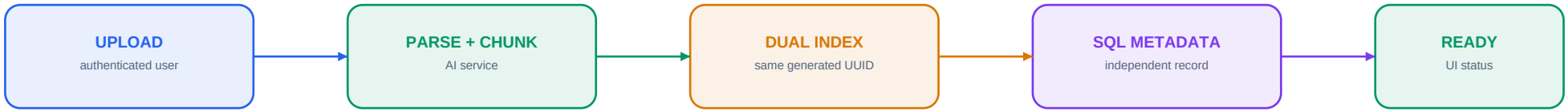
- documents collection: 384-dim BGE vectors, text and chunk metadata.
- conversation_history: masked user/assistant exchange vectors by session_id.
- External server first; local-disk then in-memory fallback if unavailable.
- Optimized for cosine similarity, not relational ownership constraints.

LEXICAL STORE

Elasticsearch

- documents index stores the same chunk UUID, text and metadata.
- BM25 supports exact terms, clause labels, identifiers and section numbers.
- Optional dependency; local TF-IDF over Qdrant candidates is the fallback.
- Merged with semantic rank using Reciprocal Rank Fusion.

Ownership and synchronization contract



Current consistency gap: document deletion removes only the PostgreSQL metadata row. Qdrant and Elasticsearch chunks are not deleted, and ingested chunks do not carry a backend document ID or user ID for scoped cleanup.

End-to-end business workflows

Two main paths: conversational response and document ingestion

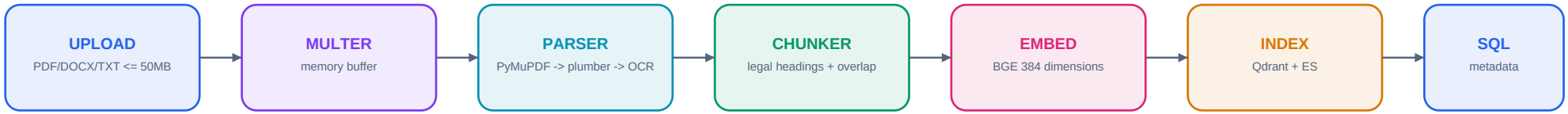
FLOW A — Authenticated chat with SSE



SSE markers: [Node: name] -> [FINAL] -> answer -> [DONE]. User message is durable before the AI call; assistant text is durable after stream completion.

Failure behavior: the frontend can fall back to non-streaming POST /api/chat; provider quota errors become a friendly final message; background fact/vector writes cannot fail the response.

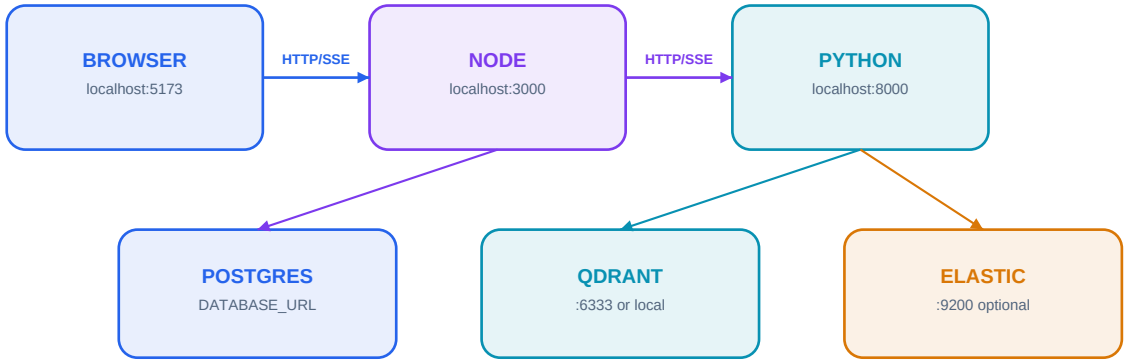
FLOW B — Upload, parse and dual-index



The SQL Document record is written after the AI ingestion attempt and records Ready or Processing Failed. This is not an atomic cross-store transaction; reconciliation is a recommended production addition.

Deployment, security and reliability design

Current deployment topology



IMPLEMENTED

Security controls

- JWT bearer token, 7-day expiry and bcrypt cost 12.
- Thread/document queries include authenticated user ownership.
- PII masking plus output leak/grounding validation around model calls.
- CORS allowlist and file MIME/size validation.

IMPLEMENTED

Reliability controls

- Research loop <= 2 passes; Self-RAG <= 2 revisions and <= 2 rewrites.
- LLM 429 wrapper performs up to four total attempts with backoff.
- Qdrant local and in-memory fallbacks; Elasticsearch TF-IDF fallback.
- Independent background memory tasks isolate non-critical failures.

Quality attribute	Current mechanism	Risk / limitation	Production recommendation
Availability	Three processes; dependency fallbacks	Single instances and local process lifecycle	Container health checks, restart policy, replicated managed stores
Scalability	Stateless HTTP services except in-process memory	Global memory dict and local Qdrant fallback	External session cache; horizontally scale services
Consistency	PostgreSQL for durable records	No transaction across SQL/Qdrant/ES	Outbox + idempotent indexer + reconciliation worker
Observability	OpenTelemetry, Prometheus, structured logs	No alert rules or dashboard defined	Trace correlation IDs, SLOs, alerts and dashboards

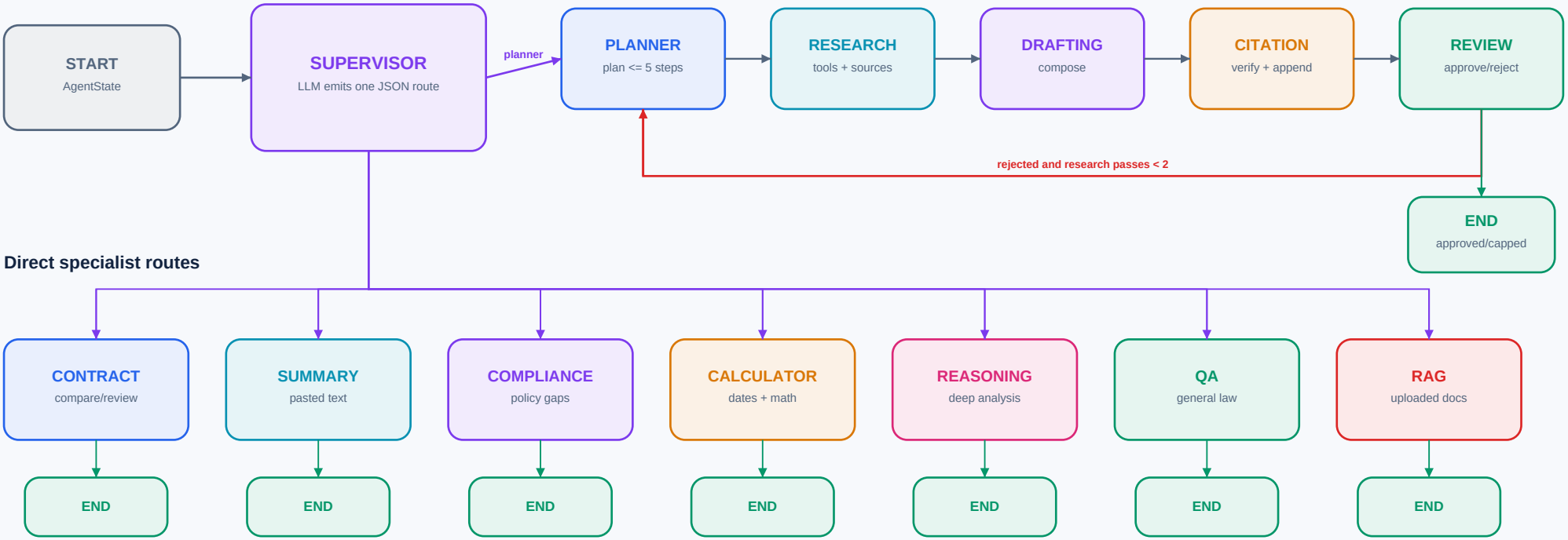
Code basis: start.sh; ai_service/docker-compose.yml; middleware/auth.js; observability

Code-level package and component map

Runtime	Module	Primary responsibility	Key collaborators	Contract
React	AuthContext / App	JWT state, profile load and protected routes	AuthPage, WorkspacePage	Bearer token + /auth/me
React	WorkspacePage	Threads, documents, chat and SSE parsing	Express routes	JSON + text/event-stream
Node	routes/auth	Register, login and profile	Prisma, bcrypt, JWT	Auth API
Node	routes/threads	Owned thread CRUD and message history	Prisma	Thread API
Node	routes/chat	Persist/proxy stream/persist	FastAPI, Prisma	Chat API + SSE
Node	routes/documents	Validate upload, proxy ingest, store metadata	Multer, FastAPI, Prisma	Multipart + JSON
Python	ChatService	Mask, memory enrichment, graph, restore	Guardrails, LangGraph	chat / stream_chat
Python	graph	Supervisor and specialist state machine	13 agent nodes	AgentState deltas
Python	rag	Ingest, hybrid search, Self-RAG and citation	Qdrant, ES, embeddings	chunks and grounded answer
Python	memory	Window, durable facts and semantic history	Postgres, Qdrant	session_id scoped
Python	masking / guards	PII replacement and response checks	Presidio, spaCy	MaskingState map

Dependency direction: UI -> Express gateway -> FastAPI orchestration -> providers and stores. The browser has no database or provider credentials; the AI service does not own user authentication or the durable message ledger.

LangGraph supervisor and specialist state machine



STATE

AgentState merge rules

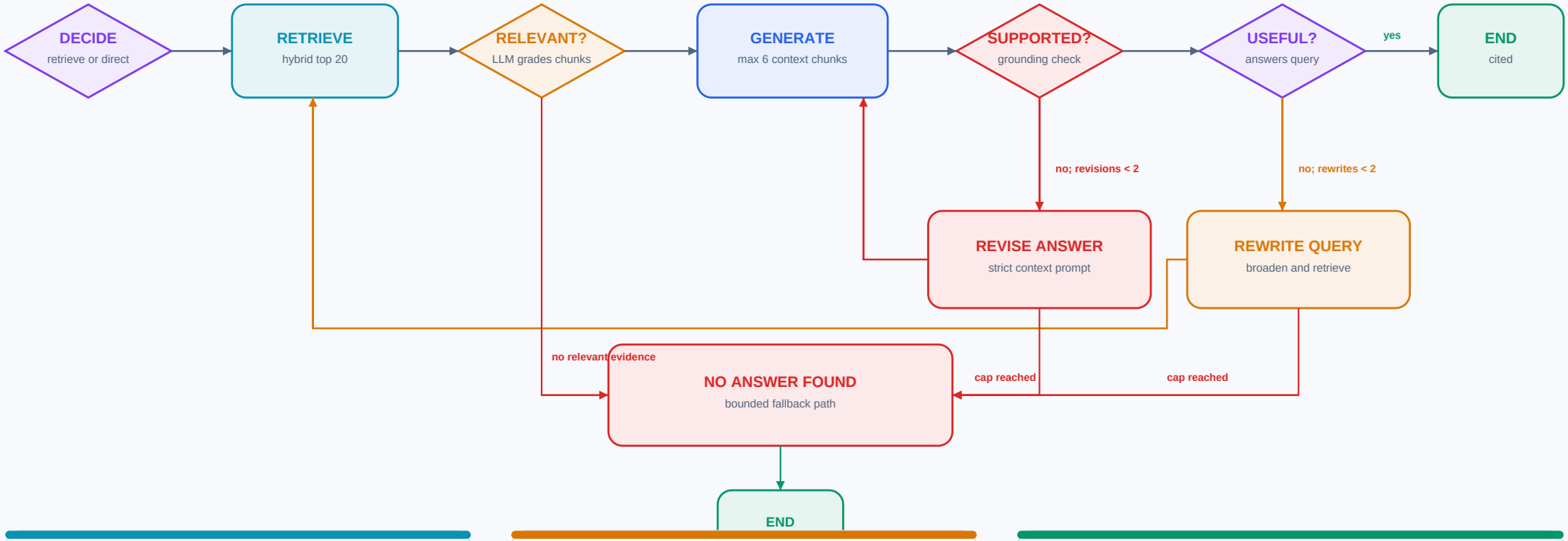
- messages uses Annotated[List[BaseMessage], operator.add] and appends deltas.
- Routing, research and specialist output fields replace previous values.
- TypedDict(total=False) permits nodes to return only their changed fields.

BOUNDED

Loop safety

- Reviewer may route to planner only while research_data count is below 2.
- At the cap, rejected output terminates rather than looping indefinitely.
- All specialist agents are terminal; direct supervisor FINISH uses a direct LLM answer.

Self-RAG retrieval and answer-repair state machine



EXACT

Retrieval numbers

- Qdrant semantic search: up to 20 candidates.
- Elasticsearch BM25: up to 20 candidates.
- RRF merge: maximum 20; reranker: top 6.

BOUNDS

Maximum normal work

- Initial retrieval plus up to 2 query rewrites.
- Initial answer plus up to 2 answer revisions.
- No exhausted check can form an infinite cycle.

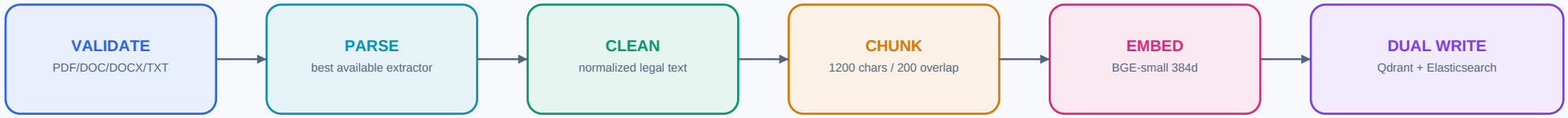
GROUNDING

Answer contract

- Use only retrieved evidence for document answers.
- Run hallucination, usefulness and citation checks.
- Return a controlled fallback after exhausted checks.

Document ingestion and hybrid retrieval algorithms

Ingestion pipeline



FALLBACK

PDF parser waterfall

1. PyMuPDF: fast page text extraction.
2. pdfplumber: complex layout and tables.
3. PaddleOCR: scanned/image-only pages.

CHUNK

Legal-aware chunk metadata

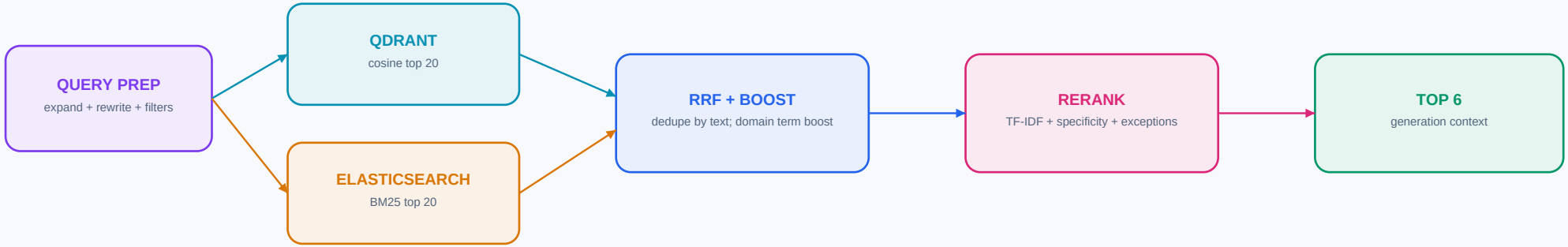
- Section number/title and document name.
- chunk_id, index, previous and next IDs.
- Large clauses split; headings copied to continuations.

CONSISTENCY

Shared index identity

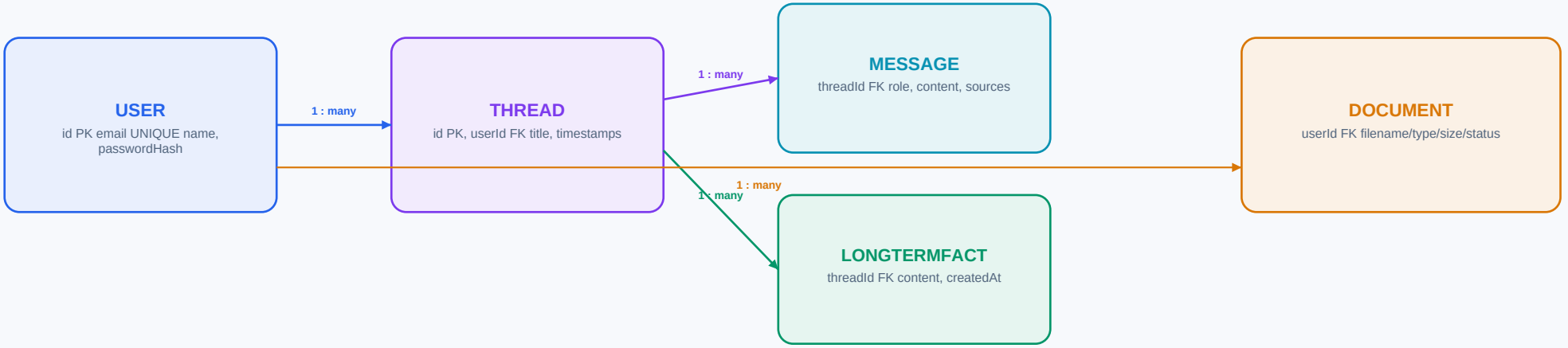
- One generated UUID is used for the Qdrant point and ES document.
- Qdrant payload and ES source contain text plus metadata.
- Index errors are logged independently.

Query-time hybrid retrieval



If Elasticsearch is unavailable, local TF-IDF scores the Qdrant candidates. If the external Qdrant server is unavailable, the client falls back to local disk and then in-memory storage.

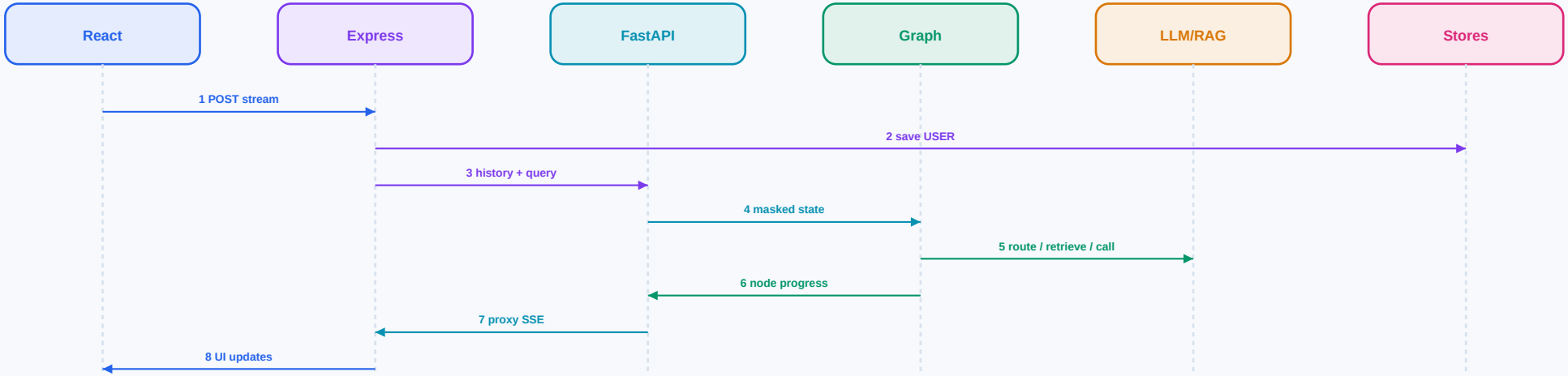
Relational model and API contracts



Cascade rules: deleting a Thread removes Message and LongTermFact rows; deleting a User cascades relational children.

Gateway endpoint	Verb	Auth	Request / behavior	Response
/api/auth/register	POST	No	name, email, password; bcrypt hash	JWT token
/api/auth/login	POST	No	email, password	JWT token
/api/auth/me	GET	Yes	verified token	profile
/api/threads	GET/POST	Yes	list owned / create	thread records
/api/threads/:id	GET/PATCH/DELETE	Yes	ownership check	history / renamed / deleted
/api/chat/stream	POST	Yes	threadId, message; save and proxy	SSE markers and answer
/api/chat	POST	Yes	non-streaming fallback	assistant Message
/api/documents	GET	Yes	list owned metadata	Document[]
/api/documents/upload-and-ingest	POST	Yes	multipart file <= 50 MB	metadata + ingest stats
/api/documents/:id	DELETE	Yes	metadata ownership check	SQL deletion confirmation

Runtime sequence, configuration and engineering decisions



ENV

Runtime configuration

- DEFAULT_LLM_PROVIDER defaults to groq.
- QDRANT_URL=:6333; ELASTICSEARCH_URL=:9200.
- DATABASE_URL and provider API keys are environment-owned.

MODELS

Provider abstraction

- Groq Llama 3.1 8B, GPT-4o-mini, Gemini 2.5 Flash, Claude 3 Haiku.
- Shared factory and rate-limit wrapper.
- Provider may be selected per AI chat request.

BOUNDS

Memory limits

- Sliding window: 20 messages.
- Long-term facts: maximum 25 per thread.
- Vector prompt recall: top 2 related exchanges.

Priority	Current issue	Design change	Acceptance signal
P0	Deleting a document leaves vector and lexical chunks	Persist document_id/user_id in chunk metadata; transactional outbox deletion	No retrieval after delete; replay-safe worker
P0	Uploaded document collection is not tenant-filtered	Enforce user_id/document_id filters at ingest and retrieval	Cross-user retrieval test stays empty
P1	Cross-store ingestion is best-effort, not atomic	Ingestion job table, idempotency key and reconciliation states	Retry produces one logical document
P1	In-memory conversation cache is process-local	Use Redis or rely entirely on durable backend history	Multiple AI replicas return identical context
P2	Local fallback stores may hide infrastructure failure	Environment-specific fallback policy and health readiness	Production readiness fails closed